S4: Training guide (descriptive and visual) to the standardised annotation attributes for identifying stranded cetaceans in satellite imagery

The following training guide aims to aid the strandings community to assess the certainty of stranded cetaceans detected in VHR optical satellite imagery. The document provides (1) a descriptive guide and (2) a visual guide for the standardised (required and optional) attributes for data annotation. The guide should be viewed by observers ahead of reviewing VHR satellite imagery for stranded cetaceans, to familiarise themselves with examples of stranded cetaceans and confounding features.

As satellite imagery is a novel tool for the strandings community, the standardised values for annotating features have been developed based on existing knowledge of strandings and the current state of the field. This is not a static document, if a value required is not available in the attributes, users are requested to add and share any updates with the corresponding author to incorporate into the main template. New proposed values should use lower case and '_' instead of space for compatibility with coding languages. This template is a baseline and intended to develop with the field.

1. Descriptive guide

The descriptive guide provides a definition for each attribute and its associated values, for the standardised metadata template proposed to the strandings community, when assessing VHR optical satellite imagery for stranded cetaceans.

Table 1. List of attributes associated with each strandings from space annotation. Required attributes (as informed by those in attendance at the Society of Marine Mammalogy Conference 2025 'Satellites to study live and stranded cetaceans: Establishing best practice' workshop, are those considered essential to include in published datasets as a minimum) are bolded, else the additional attributes not bolded are considered optional by the 'satellites to study whales' community. Attributes marked with '' are those specific to features identified as stranded_carcass, or floating_carcass only. Most attribute values are standardised drop-down menus with exception of 'id', 'observer', 'location', 'img_cat_id', 'img_date', 'img_time', '*length', '*width', and 'comments', these are user defined, and 'longitude' and 'latitude', which are calculated post-annotation. Details for environmental variables require the conditions at the point feature / annotation specifically, and not the whole image.*

Attribute	**Description**	**Unit**	**Value**	**Format**
id	A unique numerical value assigned to each annotation (marked point).	None	Numbers	*e.g. 0, 1, 2, 3…*
observer	Name of person reviewing the image (user to define the names of each person in the team of observers for a given project. Numerical values can be	None	Letters or Numbers	*e.g.* penny_clarke *or* *1*

Attribute	Description	Unit	Value	Format
	assigned to preserve observer anonymity).			
location	Name of the location where the satellite image was captured (user to define, use lower case and '_' instead of space for compatibility with coding languages)	None	Letters	*e.g.* golfo_de_penas
satellite	Name of the satellite that captured the image ('satellite' is a list of commercial VHR satellites (specifically those, which are internationally accessible), including those decommissioned, currently in orbit, or projected for launch in the near future).	None	Letters and numbers	airbus_pleiades airbus_pleiades_neo albedo_clarity_1 maxar_geoeye_1 maxar_quickbird maxar_worldview_1 maxar_worldview_2 maxar_worldview_3 maxar_worldview_4 maxar_worldview_legion planet_skysat planet_pelican
img_cat_id	Unique identification that the satellite imagery provider assigns to each image. With Maxar, this corresponds to the catalog ID.	None	Letters and numbers	*e.g.* 10400ED2959
img_date	Date the image was captured.	Year, month, day (ISO8601 format)	Numbers	(UTC) YYYY-MM-DD

Attribute	Description	Unit	Value	Format
img_time	Time the image was captured.	Hours and minutes (ISO8601 format)	Letters and numbers	(UTC) hh:mm:ssz
gsd_m	The ground sampling distance (is the distance on the ground represented by a single pixel in a satellite image), which in QGIS can be found in the metadata, by right clicking on the panchromatic file and selecting 'Properties', then 'Information' and 'Pixel Size' or else by viewing the .imd or .xml file.	meter	Decimal numbers	0.1 0.15 0.2 0.3 0.31 0.4 0.46 0.5 0.6 0.65 0.7 0.75 0.8 0.9 1
prod_type	The product type indicates the level of pre-processing applied to an image. 'prod_type' is a list of commercial VHR satellites operators, specifically those, which are internationally accessible, and their available product types.	None	Letters and numbers	airbus_primary airbus_projected airbus_ortho albedo_photogrammetry_ready albedo_view_ready albedo_analysis_ready maxar_system-ready_(basic)_1b maxar_system-ready_stereo_(basic)_1b maxar_view-ready_(standard)_or2a maxar_view-ready_stereo_(standard)_or2a maxar_view-ready_(standard)_2a maxar_map-ready_(ortho)_1:12,000 planet_skysat_basic_scene

Attribute	Description	Unit	Value	Format
				planet_skysat_ortho_scene planet_skysat_ortho_collect
projection	Projection applied to the image to remove distortion (Universal Transverse Mercator (UTM), is a series of grid zones of the world, a plane coordinate grid and can be determined for an image using https://www.dmap.co.uk/utmworld.htm.	None	Letters and numbers	*e.g.* epsg:32718_wgs84/utm_zone_18s
pansharpen	The algorithm used to perform pansharpening of an image (format includes available algorithms in QGIS OrfeoToolbox (OTB) and GDAL. For example, OTB algorithms are either 'rcs', 'lmvm', or 'bayes', and interpolation is either 'bco', 'nn' or 'linear') (user defined alternatives can be added to value mapping where necessary).	None	Letters	gdal_average gdal_bilinear gdal_cubic gdal_cubic_spline gdal_lanczos_windowed_sinc gdal_nearest_neighbour otb_bayes_bco otb_bayes_nn otb_bayes_linear otb_rcs_bco otb_rcs_nn otb_rcs_linear otb_lmvm_bco otb_lmvm_nn otb_lmvm_linear
band_combi	The combination of bands used to visualise an image (spectral bands are the ranges of	None	Numbers	3,2,1 1,2,3 4,3,1 4,3,2 5,3,2

Attribute	Description	Unit	Value	Format
	wavelengths of light that are captured by a sensor, assigned to bands). Combinations of bands can alter image visualisation. To visualise as the eye would expect, typical band combinations for Maxar imagery are R = 3, G = 2, B = 1 (four band image) and R = 5, G = 3, B = 2 (eight band image).			6,4,3
feature	The type of feature a detection is. This includes the ability to annotate confounding features, which can be useful in creating multi-class training datasets (see definitions below).	None	Letters	stranded_carcass floating_carcass live_cetacean boat marine_litter organic_beach_debris rock
*deco_phase	The level of decomposition of a carcass (see definitions below).	None	Letters	fresh_alive fresh_dead bloating_and_floating active_decomposition passive_decomposition dry_remains unknown
***certainty**	Certainty of an observer that a detection is the assigned feature	None	Letters and numbers	definite_(90-100%) likely_(70-89%) possible_(50-69%)

Attribute	Description	Unit	Value	Format
	(species or higher taxonomic level) **Definite**: 90-100% confidence a detection is the defined feature (almost no doubt) **Likely**: 70-89% confidence a detection is the defined feature (high likelihood but not sure, for example, when in a mass it is challenging to determine individuals, however based on morphology and size you assign as two carcasses, but you are not sure) **Possible**: 50-69% confidence a detection is the defined feature (hard to tell, perhaps because there is marine debris on the beach with a similar appearance or an individual is far from the mass but has a similar appearance).			
*species	The species or higher taxonomic level of the feature detection (*e.g.*	None	Letters	*e.g.* unidentified_whale unidentified_dolphin unidentified_small_whale

Attribute	Description	Unit	Value	Format
	'sei_whale'), else when unknown assign higher taxonomic level (*e.g.* 'large_whale' or 'unidentified_whale'). The list of species uses the Society of Marine Mammalogy (SMM) (Committee on Taxonomy, 2024) naming conventions, to ensure consistent terminology throughout the marine mammal community, with addition of instances where species is unknown.			unidentified_large_baleen_whale unidentified_large_whale (… remaining list is the SMM naming conventions for all species *e.g.* bowhead_whale)
*colour	The most dominant colour of the detected carcass (during decomposition the carcass colour may be a mixture of colours, assign the most dominant by percentage cover).	None	Letters	black brown cream dark_grey light_grey orange pink red white
*body_shap	Overall shape of the body excluding fluke and flippers. Select the most suited or 'undefinable', especially in late stages of decomposition.	None	Letters	ellipsoid streamline irregular_ellipsoid irregular_ streamline undefinable

Attribute	Description	Unit	Value	Format
*length	Maximum visible length between the tip of the head and the fluke with values ranging from calf size to maximum adult length.	Meters	Numbers	*e.g.* 12.4
*width	Maximum visible width measured at the widest part of the body and perpendicular to the body length.	Meters	Numbers	*e.g.* 2.0
*flippers	Are the flippers (forelimb used to turn, break and stabilise a cetacean while swimming) visible?	None	Letters	yes no maybe
*fluke	Is the fluke (tail used to propel a cetacean forward) visible?	None	Letters	yes no maybe
*slick/blo	Is a slick or blood visible? A slick is a waxy grey liquid released when cetacean decomposes (the oil contained within cetacean blubber gets converted to adiopocere, a wax like substance formed when body lipid-tissues break down. It can cover wide areas around the remains. This	None	Letters	yes no maybe

Attribute	Description	Unit	Value	Format
	can temporarily prohibit the growth of nearby plants, so reduced or dead vegetation nearby and grey colouration surrounding a carcass may be seen from space.			
*group	The number of carcasses within 1 km of a detected feature. This should be per annotation and can be an estimate, especially when in a mass.	None	Letters and numbers	Individual 2_to_5 6_to_10 11_to_50 51_to_100 101_to_200 201_to_500 501+
cloud_cov	Cloud cover for the area within the vicinity of the detection, using the aviation system: 'skc_(sky_clear)' = 0/8th 'few_(traces)' = 1-2/8th 'sct_(scattered)' = 3-4/8th 'bkn_(broken)' = 5-7/8th 'ovc_(overcast)' = 8/8th	Okta	Letters	skc_(sky_clear) few_(traces) sct_(scattered) bkn_(broken) ovc_(overcast)
cloud_thic	Cloud thickness for the clouds present within the vicinity of the detection:	None	Letters	none thin medium_thin

Attribute	Description	Unit	Value	Format
	Thin (can see fairly well through the clouds) Medium thin (can see through but no clear view of the sea) Thick (cannot see through)			thick
environmen	The abiotic environment that a detection is found in.	None	Letters	bedrock cobble_beach estuary_mud_flat gravel_beach man_made_structure mangrove marsh_brackish marsh_freshwater nearshore_waters_<1km offshore_waters_>1km sand_beach_dark sand_beach_light shell_beach vegetated_trees vegetated_wetland
img_bright	The brightness of the image within the vicinity of the detection.	None	Letters	bright moderate dark
shade	The level of shade from the sun, full, partial or none at the	None	Letters	full_shade partial_shade no_shade

Attribute	Description	Unit	Value	Format
	point of the detection.			
ice_cover	Relevant to ice covered regions only, the level of ice cover at the point of the detection.	None	Letters	none low medium dense very-dense
comments	Any other comment the observer would like to make about the specific detection.	None	Letters	*e.g.* Could be dry remains of carcass or wood from a tree
longitude	Longitude of the carcass.	Decimal degree	Numbers	*e.g.* 12.70
latitude	Latitude of the carcass.	Decimal degree	Numbers	*e.g.* 67.50

Recommended file naming convention for sharing exported annotation metadata .csv files and for application of automated clustering code to compare counts (https://github.com/PennyJClarke/strandings_from_space): 'location_satellite_image-catalog-id_image-date_image-time_spatial-resolution_observer.shp'. All entries should use lower case and '_' instead of space for compatibility with coding languages.

Feature ID (id)

The feature 'id' is a unique numerical value assigned to each marked point. It is recommended this value is entered manually for each new feature, starting at 0, assign each new feature an incremental numerical value, for example, 1, 2, 3, 4, etc. The tool used to export image chips in the final step of the process, numbers from zero, and therefore, will correspond with the original feature 'id' assigned. There is an option to make this field populate automatically with the next incremental value, however, this is not recommended, as anytime a change is made this results in automatic reassignment to a new 'id' value (for example moving a point's position).

Image Observer (observer)

This field is project specific, allowing the user to define the 'Value' and 'Description' with the names of each person in the team of observers for a given project. Ensure to use lower case and hyphens (_) instead of space, for example, penny_clarke. Alternatively, observers can be assigned a numeric value instead of a name for anonymity. Guidance on setting up dropdown entries with observers names or auto filling an attribute table column with a given name can be found in the 'S2_QGIS_Strandings_from_Space_Pipeline' document.

Image location (location)

Location is a user defined field to detail the name of the location where the satellite image was captured (lower case and '_' instead of space for compatibility with coding languages). The chosen name should be descriptive so that another user could identify the general location the image was collected.

***The following attributes associated with information of the satellite image can be found in the satellite image metadata file. N*avigate to the panchromatic or multispectral image folder in your directory, double click the .XML or .IMD file in the folder to open and view the image metadata. Precise details with how to locate the information within the metadata file can be found in the 'S2_QGIS_Strandings_from_Space_Pipeline' document.**

Satellite sensor (satellite)

Satellite is a list of the VHR optical satellites sensors for the main commercial operators, Airbus, Albedo, Maxar, and Planet. Users should select the satellite sensor used to capture the image being reviewed. The list includes decommissioned sensors, those currently in orbit, or projected for launch in the near future.

Image catalogue ID (img_cat_id)

Image catalogue ID is a user defined field that should correspond to the unique identification number or series of letters and numbers that the satellite imagery provider assigns to each image. Ensure to enter the ID precisely to include any instances where an ID contains text separated by hyphens or dashes.

Image collection date (img_date)

The image collection date is a user defined field detailing the date image was captured. The date should be recorded as year, month, day (YYYYMMDD) coordinated universal time (UTC, worldwide standard to ensure timekeeping), in accordance with the ISO8601 format.

Image collection time (img_time)

The image collection time is a user defined field detailing the time image was captured. The time should be recorded as hours, minutes, and seconds (hh:mm:ssZ) UTC, in accordance with the ISO8601 format ('T' is removed from the beginning of Thh:mm:ss as per ISO8601 guidelines, as this is not necessary to include unless for separation of date and time when communicated together).

Ground sampling distance (gsd_m)

The ground sampling distance is the distance on the ground represented by a single pixel in a satellite image. The distance is given in meters to ensure that users could apply this standard to annotate more coarse resolution imagery exceeding 1 m spatial resolution, using the same unit. Values in the image metadata may be given in scientific notation, where 5.000000000000000e-01 is equivalent to 0.5 m or 50 cm.

Product Type (prod_type)

Product Type is a list of product types for the main VHR satellite imagery providers. Satellite image companies can provide images at differing levels of pre-processing, which are defined by the user at the time of ordering imagery. Pre-processing can include, projecting, pansharpening, orthorectification, and atmospheric correction of imagery. Users should select the product type associated with an image, by the image provider who supply the imagery.

Projection (projection)

Projection is a user defined entry to denote the projection applied to the image to remove distortion. When a satellite image is collected, the image is a flat representation of a spherical Earth, and the image may therefore appear distorted. By assigning a coordinate system that is most relevant to the image area through projection / re-projection, the image representation will show a more accurate flat two-dimensional representation of the Earth. A projection is a mathematical transformation of a coordinate system. If the image is projected it will be detailed within the image metadata, else for images without a projection, users can assign the most appropriate projection. Universal Transverse Mercator (UTM), is a series of grid zones of the world, a plane coordinate grid and can be determined for an image using https://www.dmap.co.uk/utmworld.htm (where images overlap zones, best practice would be to choose the UTM zone containing the majority of image). To comply with coding languages entries should be made as all lowercase with hyphens used instead of space, for example, 'epsg:32718_wgs84/utm_zone_18s'.

Pansharpening algorithm (pansharpen)

Pansharpening algorithm is a list of algorithms available in QGIS that can be applied to pansharpen an image. A panchromatic image will have a higher spatial resolution than a multispectral image, as it captures a wider range of light in a single band, allowing it to be significantly sharper. For example, Maxar's WorldView 3 sensor, produces a panchromatic image with a spatial resolution of 30 cm, and a multispectral image with a spatial resolution of 1.24 m. To analyse a multispectral image, it can be useful to enhance its spatial resolution to the higher spatial resolution of the panchromatic image, through the process of pansharpening. Pansharpening takes the panchromatic and multispectral images, and where the two rasters fully overlap, fuses them together to produce a multispectral image with the higher spatial resolution of the panchromatic image. Pansharpening is useful to studying cetacean strandings from space as both spectral and spatial richness is needed to discriminate individual carcass with confidence. The list details possible algorithms in QGIS using a plugin called Orfeo Toolbox (OTB) (BundleToPerfectSensor) and GDAL. Both tools offer a series of algorithms to pansharpen imagery, for example, one of OTB's algorithm options is 'bayes' algorithm with 'bicubic interpolation', if users applied this algorithm, they would select 'otb-bayes-bco'.

Band combination (band_combi)

Band combination is a series of common bands combinations used to review imagery, *e.g.*, RGB or false colour composite. The band combinations to visualise multispectral imagery can be ascertained by reviewing the satellite sensor specification, either online from the satellite image provider, or within the image metadata file. Typically for Maxar imagery, to view the imagery as our eye would naturally see, four band imagery is R = 3, G = 2, B = 1 (R is Red, G is Green and B is blue bands) and eight band imagery is R = 5, G = 3, B = 2. The band combination R = 6, G = 4, B = 3 (in eight band imagery) can be a powerful combination for strandings monitoring, to discriminate stranded whales from confounding features. Using this band combination, stranded whales appear unique, when compared with the red of vegetation (Fretwell et al., 2019).

Feature (Feature)

Feature is a category assigned to each marked point to describe what a feature is, the categories are:

1) stranded_cetacean: a live or dead cetacean located onshore. A stranding is a cetacean that falters ashore, debilitated, or is in an environment incompatible with its natural survival (*e.g.*, washed ashore on a beach); if the cessation of life has occurred prior to deposition, the cetacean is considered beachcast (Clarke et al., 2021). As it is not possible to determine from space whether a feature washed ashore is alive or dead (beachcast) at the time of stranding, all carcasses considered stranded should be assigned stranded_carcass. Partially beached animals within the surf zone should be defined, stranded_cetacean.

2) floating_carcass: cetaceans that die at sea (or those who have stranded and since been washed offshore) may float due to decomposition gasses keeping the carcass at the surface. In these instances, a dead cetacean located in the near shore zone or offshore in the open ocean zone is considered a floating_carcass.

3) live_cetacean: a live (free-swimming) cetacean. While reviewing imagery for stranded cetaceans it is possible that open water may be visible within the imagery and live (free-swimming) cetaceans may be detected, these can be annotated as a basic annotation to denote the location, for later incorporation into live cetacean datasets using Cubaynes et al. (2023) recommended attributes.

4) boat: small or large boats located on the shore, or in the near shore and offshore zones if identifying floating_carcass and live_cetacean.

5) marine_litter: any item not naturally found in the marine environment, for example, plastic and lost, discarded, abandoned or ghost fishing gear, which can exhibit a similar appearance to a stranded cetacean feature from space.

6) organic_beach_debris: sometimes termed wrack/strandline debris/organic marine debris and includes seaweed, algae, seagrass and woody materials such as tree trunks, branches and bark, which may have a similar appearance to a stranded cetacean feature from space.

8) rock: rock that has a similar appearance to a stranded cetacean feature.

We recommend annotating confounding features, features that are visually like a cetacean, for example, algae, trees, boats, and rocks. Confounding features should only be assigned when there is certainty. By annotating confounding features, these annotations could be incorporated into training datasets for multi-class machine learning models, to better train models at detecting stranded cetaceans correctly.

The following 'Fields' with an asterix are those relevant to cetacean features only.

Phase of Decomposition (*deco_phas)

Upon stranding, a cetacean unless recovered will undergo decomposition, and the level of decomposition can be attributed to a phase as it progresses. Stranding networks working on the ground have well established decomposition levels definitions used in their response programmes and records. Standardisation in use of terminology between stranding network experts working on the ground and remote sensing experts studying strandings from space, is essential to working collaboratively and cross-disciplinary. Ideally, decomposition level definitions used by stranding networks would be incorporated into space applications. However, stranding network ground-based definitions, for example, for the United Kingdom (UK) criteria (Law et al., 2006), are driven by details that require close observation of an animal, and are more driven by pathology and the internal structure and integrity of the organs. As strandings detection from space uses remote monitoring with limited detail, the definitions for phases of decomposition need to focus more on externa. As such we propose standardised phases of decomposition definitions established through time-lapse imagery as part of the largest mass mortality event of baleen whales in the world, in the Chilean Patagonia, 2015 (pers. comms. McConnell, K., 2025, informed by Payne (1965) and Mądra et al. (2015)). For each decomposition phase definition, similarity to stranding response team definitions from the UK are detailed where appropriate, and in addition, comparison and incorporation of the decomposition condition categories (DCC) from the Agreement for the Conservation of Cetaceans of the Baltic Sea, Mediterranean Sea and Contiguous Atlantic Area (ACCOBAMS) and Agreement on the Conservation of Small Cetaceans of the Baltic, North East Atlantic, Irish and North Seas (ASCOBANS) best practice on cetacean post mortem (IJsseldijk et al., 2019). Regional differences in timescales for each phase of decomposition may apply, for example, tropical / warmer climates will experience faster rates of decomposition when compared with temperate and again cooler polar climates. In most instances these phases of decomposition may not be attributable without ground truth, knowledge of the timing of the event (stranding network reports) and local environment etc. **In these initial stages of satellites application to study strandings, phases of decomposition should only be assigned where ground truth data is available**. If one of the following phases of decomposition cannot be assigned with confidence, then assign as 'unknown'. It is unlikely 'fresh_alive' will be captured by satellites at present, as current technological advancements do not support emergency response to live mass strandings, due to time delays in image collection (Clarke et al., 2021), though as technology improves this could become possible:

1) fresh_alive: stranded alive (UK: condition code 1, DCC code 1).

2) fresh_dead: Normal appearance, carcass not bloated, tongue and penis not protruded; blubber white and occasionally tinged with blood. Typically, during the first 24-48 hours after death, decomposition processes immediately begin acting on and, in the carcass, (timings may vary across the globe and local knowledge should be applied to a region) (UK: condition code 2a, DCC code 1 & 2, figure 1).

Figure 1: 'fresh_dead' sei whale (Balaenoptera_borealis), exhibiting normal appearance and no bloating, captured in the Golfo de Penas, Chile, © Katie McConnell 2025

3) bloating_and_floating: Bloating evident, with tongue and penis often distended; skin may be visibly cracked and blubber is blood-tinged and oily. Fermentation of the internal organs causes the whale carcass to swell and float during high tides. Floating carcasses can be transported great distances and so may be found offshore (UK: condition code 2b / 3, DCC code 3, figure 2).

Figure 2: 'bloating_and_floating' sei whale with evident bloating (a, b), distended tongue (a) and blood-tinged blubber (a, b), captured in the Golfo de Penas, Chile, © Katie McConnell 2025

4) active_decomposition: 2-3 weeks after death, the carcass may be open, beginning rapid tissue loss via decomposition, microbes, maggots and scavengers, or the carcass may be intact but collapsed, skin sloughing, epidermis may be largely missing, exposing underlying blubber. As lipid-based tissues, like blubber are converted to adipocere, a waxy grey liquid may be released and can cover wide areas around the remains. This can temporarily prohibit the growth of nearby plants. This may be visible from space, directly as a grey colouration surrounding a carcass during this phase, or indirectly due to reduced vegetation cover/death of vegetation nearby (UK: condition code 4, DCC code 4, figure 3).

Figure 3: (a, b) A sei whale in 'active_decomposition' with rapid tissue loss and visible adipocere (waxy grey liquid), and (c) resulting prohibited growth of vegetation as a result of adipocere, captured in the Golfo de Penas, Chile, © Katie McConnell 2025. (d) Vegetation death associated with the conversion of lipid-based tissues to adipocere from a sperm whale in 'active_decomposition', captured in the Humber estuary, UK © Penny Clarke 2025.

The release of oils associated with this phase of decomposition could result in high reflectivity in satellite imagery, making the feature appear bright, even when visibly on the ground the carcass is predominantly dark in colouration (red/black) (figure 4).

Figure 4: (a) A sperm whale in ‘active_decomposition’ captured from the ground in the Humber estuary, UK © Penny Clarke 2025. The carcass is intact but collapsed, with skin sloughing and the epidermis largely missing exposing the underlying blubber. Lipid-tissue conversion to adipocere is visible by the presence of a waxy grey liquid and death of the surrounding vegetation. From the ground the remaining skin of the carcass appears black and cracked due to sun exposure and decomposition, and the exposed blubber is blood tinged, giving the carcass an overall black and red colouration. (b) The same carcass captured in satellite imagery exhibits a bright white colouration, potentially as a result of the high reflectivity of adipocere © Maxar Technologies 2025.

5) passive_decomposition: Once the maggots leave the remains, the few remaining tissues are those that are more tough. Skin may be draped over skeletal remains; any remaining tissues are desiccated. Organs will have partially or totally disappeared. During the next year or more, storms and scavengers will gradually clean the remains from exposed skeletons (UK: condition code 5, DCC code 5, figure 5).

Figure 5: (a, b) A sei whale in ‘passive_decomposition’ with few remaining desiccated *tissues draped over skeletal remains, captured in the Golfo de Penas, Chile, © Katie McConnell 2025.*

6) dry_remains: For many years afterwards the bones from cleaned skeletons are scattered and eventually buried or disintegrated (UK: condition code 5, DCC code 5, figure 6).

Figure 6: (a, b, c) 'dry_remains' of a sei whale, with cleaned skeletons intact and scattered, captured in the Golfo de Penas, Chile, © Katie McConnell 2025.

7) unknown: without prior knowledge or ground truth data, the level of decomposition cannot be confirmed.

Certainty (*certainty)

Certainty is a percentage confidence assigned based on each individual observer's interpretation and evaluation of a feature. For example, how confident an individual observer is that a feature is for example, a stranded_cetacean.

Certainty can be defined within three categories:

definite_(90-100%)

likely_(70-89%)

possible_(50-69%)

Assigning a level of certainty is subjective and should be based on the observer's best knowledge. To help, the images in the visual guide provide examples of what a definite cetacean stranding feature and confounding features look like within satellite imagery (including fluke, slick, or group presence, which can increase the confidence in assigning certainty). Examples are provided from multiple satellite sensors with differing spatial resolution, supplemented with aerial or ground imagery, where possible.

In the initial phases of annotating features using this novel method, stranded cetaceans may be assigned a lower certainty, for example, 'possible_(50-69%)', particularly due to the heterogeneity of cetacean strandings as they decompose. With greater examples, our understanding and interpretation of features and confidence in assigning a certainty level, will increase. In these early stages, confidence in assigning certainty can be elevated from possible to probable or definite with ground truth data / images of known events. Hence the importance of testing this technology with pilot studies in known locations of strandings with available data on the ground, before testing in more remote locations without prior knowledge available. For example, information from a stranding network verifying the location of a stranding, can offer an observer greater confidence in assigning a feature.

Figure 7 provides an example of evaluating the certainty of a feature. The definite cetacean has been assigned definite, even in absence of ground truth data, owing to the whale like shape, size and visible fluke feature. The probable whale has no visible cetacean defining features like a fluke or flippers, though is within the expected size range. You may consider it looks similar to the possible feature, however, in this instance the feature was assigned probable as the feature was in the location of a known carcass detailed from ground data. Thus, demonstrating the importance of ground data in these early stages to increase the confidence in assigning a certainty value.

definite_(90-100%)	likely_(70-89%)	possible_(50-69%)

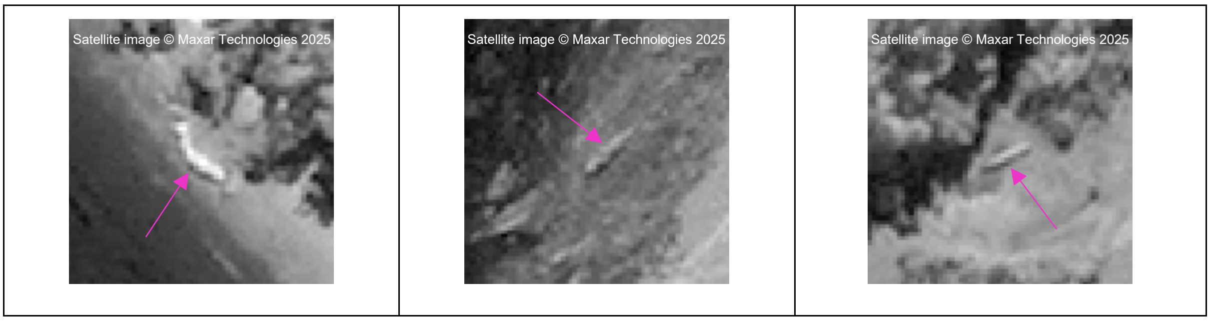


Figure 7: An example of a definite_(90-100%), likely_(70-89%) and possible_(50-69%) feature.

The environmental variable 'Fields', such as cloud cover, coastal type, and the shade conditions can all impact the confidence of an observer in assigning a certainty, hence, why the conditions at the point of the feature are noted, to greater evaluate their impact.

species (*species)

The list of species draws upon the standardised marine mammal species and subspecies defined by the Society of Marine Mammalogy (Committee on Taxonomy, 2024). The values within the template shapefile provides additional categories for stranded cetaceans identified in satellite imagery for when the species are unknown.

In the early stages of development of this tool, it is recommended the tool is applied to areas of known strandings, with the opportunity to ground truth. This will allow the field to establish examples of species in satellite imagery, providing greater capabilities to discern species in areas without prior knowledge.

Only assign a species for those you are certain of, for example, those you have ground data for, otherwise assign one of the following:

unidentified_whale

unidentified_dolphin

unidentified_small_whale

unidentified_large_baleen_whale

unidentified_large_whale

Body Colour (*colour)

Body colour details a list of colours that may be associated with a live, stranded or decomposing cetacean. Select the most appropriate colour of your feature, by determining the most dominant colour by percentage cover of the carcass body. For smaller cetaceans, mixed pixels containing mixed spectra from cetacean and non-cetacean features may influence the visible colour. Mixed pixels could result in colours atypical for a cetacean or a given species.

Body shape (*body_shap)

Body shape details a list of body shapes that may be associated with a live, stranded or decomposing cetacean. Select the most appropriate body shape for your feature. 'irregular_ellipsoid' and 'irregular_streamline' may be suitable terms to describe carcass where decomposition acting upon the carcass, causes the carcass to collapse causing irregularity in the shape. For decomposing individuals assigning a body shape may become challenging, particularly where appendages break off or in the later stages of decomposition, in these cases body shape can be assigned 'undefinable'.

Body length (*length)

Body length is a measure of the feature from the tip of the upper jaw to the notch of the tail (this may be across longitudinal (beak/fluke axis) if the cetacean is upright or if on its side across the sagittal plane from the cranial/anterior to caudal/posterior length), or the longest cross section in cases where satellite imagery is too coarse to determine jaw and tail, or where appendages are not visible (figure 8 and 9). For details on how to take a measure, review section 20 in the 'S2_QGIS_Strandings_from_Space_Pipeline' document. Measurement to be entered in meters to nearest 2 decimal points.

Figure 8: An example of measuring body length of a cetacean feature from the tip of the upper jaw to the notch of the tail.

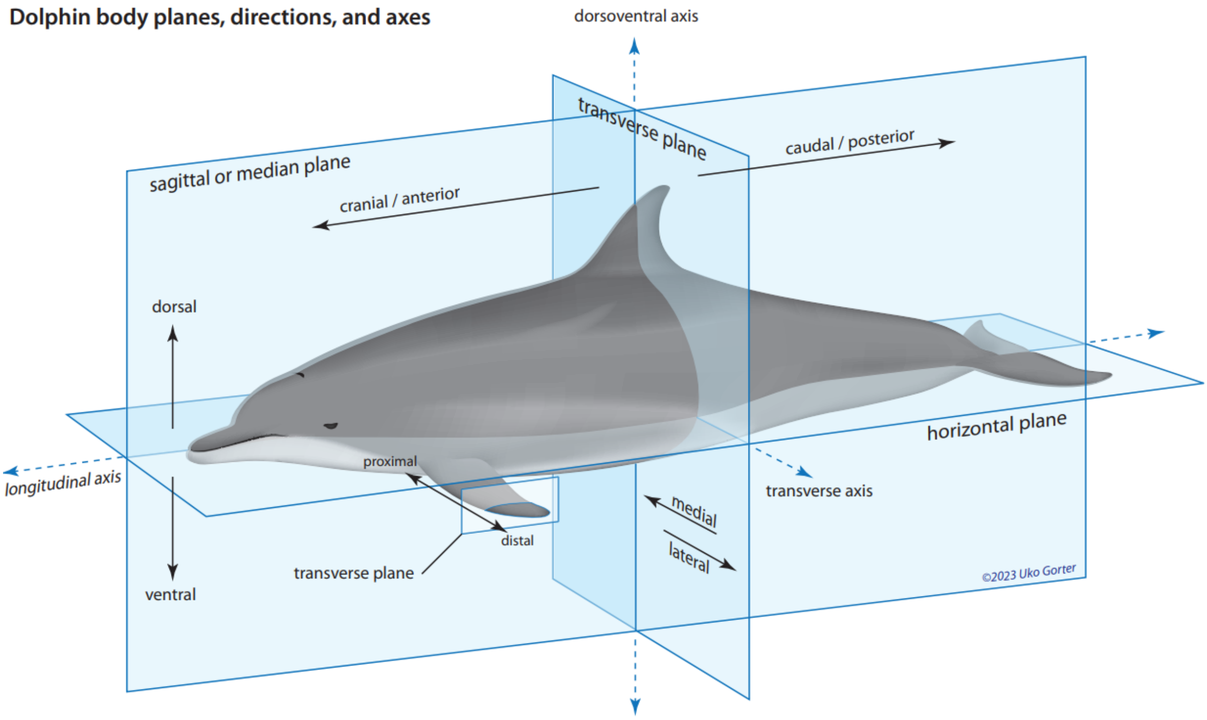


Figure 9: Positional veterinary anatomy used to define body morphometrics. For body length measurements, measure the longitudinal (beak/fluke axis) if the cetacean is upright or if on its side across the sagittal plane from the cranial/anterior to caudal/posterior length. For body width measurements, if the carcass is on its side measure across the sagittal plane from dorsal to ventral length or if the carcass is upright the transverse plane across the medial to lateral length, © Uko Gorter 2023.

Body width (*width)

Body width is a measure of the feature at its widest cross section (if the carcass is on its side this will be across the sagittal plane from dorsal to ventral length or if the carcass is upright the transverse plane across the medial to lateral length) (figure 9 and 10). For details on how to take a measure, review section 20 in the ‘S2_QGIS_Strandings_from_Space_Pipeline’ document. Measurement to be entered in meters to nearest 2 decimal points.

Figure 10: An example of measuring body width of a cetacean feature at the widest cross section.

Flippers (*flippers)

Flippers are the forelimbs used to turn, break and stabilise a cetacean while swimming. This value is a presence or absence of flippers, yes, no, or maybe. Flippers are only likely to be visible for larger stranded cetacean species, in fresh strandings of smaller cetacean species, or live cetaceans, as flesh rots and appendages break off during decomposition.

Fluke (*fluke)

Fluke is a tail used to propel a cetacean forward. This value is a presence or absence of a fluke, yes, no, or maybe. A fluke is likely only to be visible for larger stranded cetacean species, in fresh strandings of smaller cetacean species, or live cetaceans, as flesh rots and appendages break off during decomposition.

*On sandy beaches tail stocks can become buried, giving the appearance that a fluke is separate to the carcass body (figure 11). Observers should be mindful of this as it could lead to misidentification of flukes as calves.

Figure 11: An example where a tail stock has become buried by sand, giving the appearance that the fluke is separate to the carcass body.

Slick and Blood (*slick/blo)

Slick refers to adiopocere, a wax like substance formed when lipid-based tissue breaks down as whales decompose such as, blubber. The waxy grey liquid released can cover wide areas around the remains. Blood can refer to bloody directly or blood-tinged organs or blubber. This value is a presence or absence of a slick or blood, yes, no or maybe. Slicks are only likely to be visible in decomposing strandings, once the blubber is decomposing. Slicks can temporarily prohibit the growth of nearby plants, so reduced vegetation nearby or the death of vegetation and grey colouration surrounding a cetacean may be seen from space during this phase.

Group (*group)

Group is the number of individuals identified within a 1km radius of a marked point. The number of individuals does not have to be exact but a best estimate.

*****Please note: the following 'Fields' are environmental variables and requires the conditions at the point feature / annotation specifically, and not the whole image*****

Cloud Cover (cloud_cov)

Cloud cover, using the aviation industry terminology, is reported in terms of 1/8th of sky cover, otherwise known as oktas, the terms are as follows (*e.g.,* 0 oktas represent the complete absence of cloud):

'skc_(sky_clear)' = 0/8th

'few_(traces)' = 1-2/8th

'sct_(scattered)' = 3-4/8th

'bkn_(broken)' = 5-7/8th

'ovc_(overcast)' = 8/8th

Cloud Thickness (cloud_thick)

Cloud thickness (as per Cubaynes et al., 2023) should be defined as one of the following:

1) thin: can see fairly well through the cloud

2) medium_thin: can see through but no clear view of the ground

3) thick: can't see through

Environment (environmen)

Environment is a list of common environments a stranded, floating dead or live cetacean may be found, for example, a light sand beach, <1km nearshore waters, or mangroves. If the environmental context is unknown, to their best knowledge, the observer should assign an interpretation of the environment that is most applicable to the conditions visible in the satellite image at the marked point.

Image brightness (img_bright)

Image brightness can differ due to combination of factors, the suns angle, the nadir angle (the angle of the satellite sensor to the ground, 0° is directly overhead, and the greater the nadir angle, the greater the distortion), and contrast between environments, for example highly reflective ice next to absorbent ocean. Individual observers are to assign image brightness based on their interpretation of the image conditions at the marked point.

Shade (shade)

Shade is an evaluation of the light conditions at a marked point, which is influenced by the incoming UV radiation. For example, a vegetated coastline or local topography relative to the suns angle may result in a feature being in the shade. A feature may be fully or partially shaded or not in the shade.

Ice Cover (ice_cover)

Ice cover is only relevant to regions where ice is present, for example, the Polar Regions. Ice would likely mostly impact the detection of floating carcasses or live cetaceans, but in regions of dense ice, ice may arrive on shore and could impact the detection of stranded cetaceans. Ice cover is an evaluation of the density of ice present in the region at a marked point.

Comments (comments)

Comments is a space for observers to note any other information relating to a feature that may be important for data analysis.

Longitude (longitude)

Once all features are annotated, the longitude of the carcass can be extracted. For consistency and ease of comparison the recommended format is decimal degrees. Projected coordinate systems units, with a known zone and datum, are typically linear measurements, for example, meters, and will need to be transformed. Geographic coordinate systems units are typically in decimal degrees and will not require transforming. See the ‘S2_QGIS_Strandings_from_Space_Pipeline’ document for guidance on how to extract longitude using the attribute table field calculator.

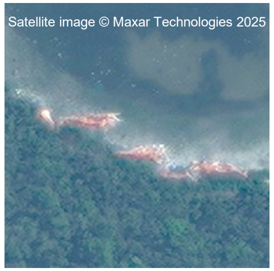
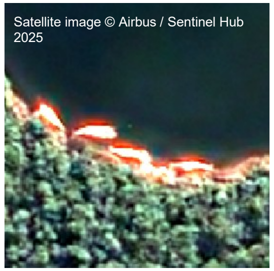
Latitude (latitude)

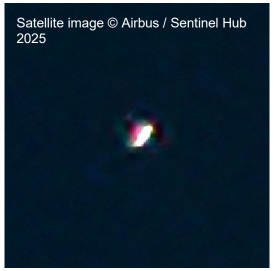
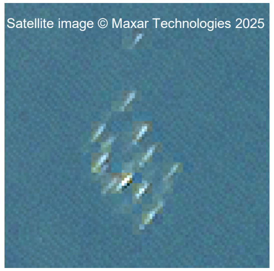
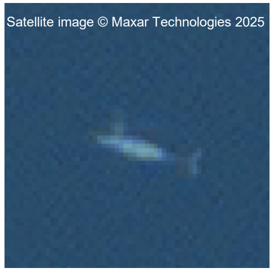
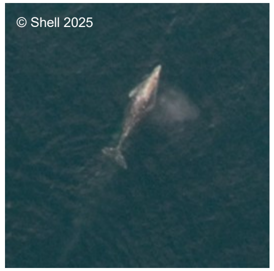
Once all features are annotated, the latitude of the carcass can be extracted. For consistency and ease of comparison the recommended format is decimal degrees. Projected coordinate systems units, with a known zone and datum, are typically linear measurements, for example, meters, and will need to be transformed. Geographic coordinate systems units are typically in decimal degrees and will not require transforming. See the ‘S2_QGIS_Strandings_from_Space_Pipeline’ document for guidance on how to extract latitude using the attribute table field calculator.

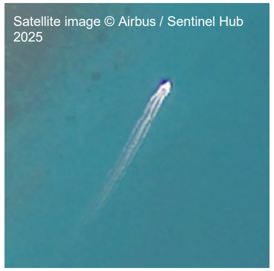
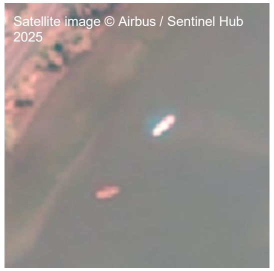
2. Visual guide

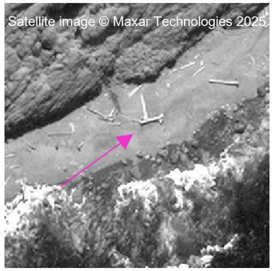
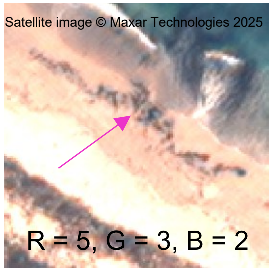
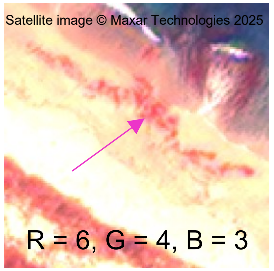
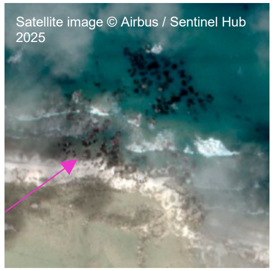
The following guide provides visual examples of the standardised attributes and associated values in VHR optical satellite imagery (0.5, 0.3 and 0.15 m spatial resolution), supplemented with aerial or ground imagery to aid the confident detection of stranded cetaceans in VHR satellite imagery.

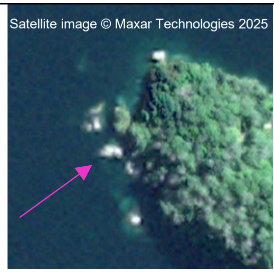
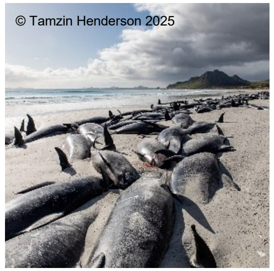
Table 2: For each standardised attribute associated with stranded carcass and floating carcass, image examples are provided for the respective standardised values, where possible from VHR optical satellite imagery across different spatial resolutions (0.5, 0.3 and 0.15 m), and supplemented with aerial and ground image examples.

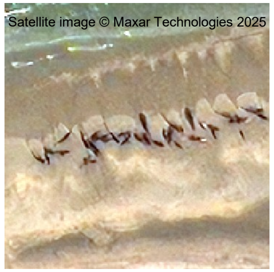
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
feature	stranded_carcass					
						
						

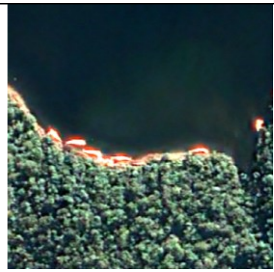
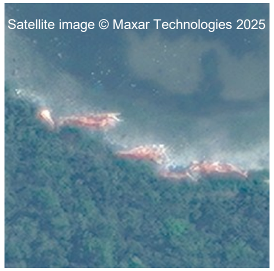
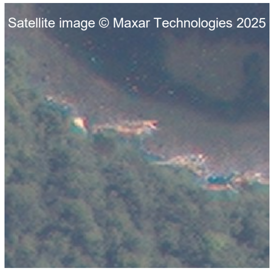
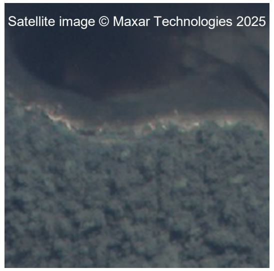
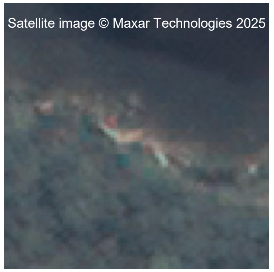
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	floating_carcass					
	live_cetacean		  	  	  	

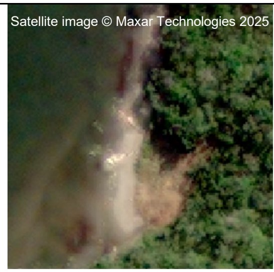
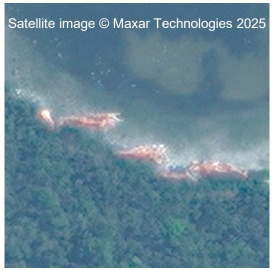
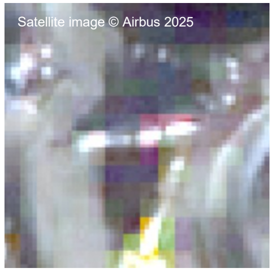
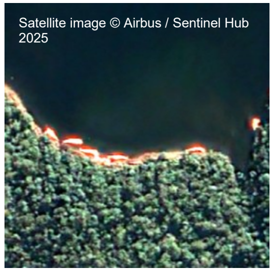
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	boat			  		
	marine_litter					

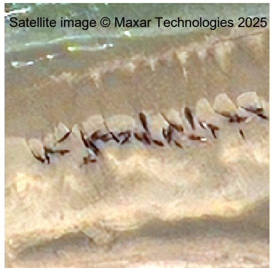
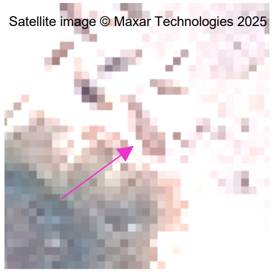
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	organic_beach_debris			  		
	rock					

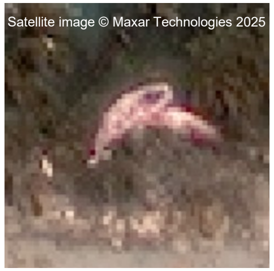
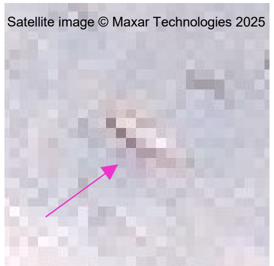
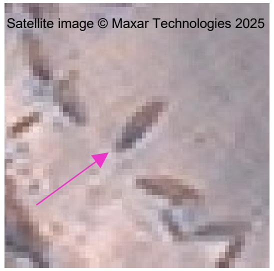
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
						
***deco_ phase**	fresh_alive					
	fresh_dead					 

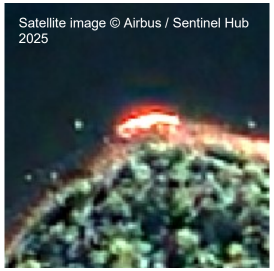
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	bloating_and_floating					  
	active_decomposition		 	 		 

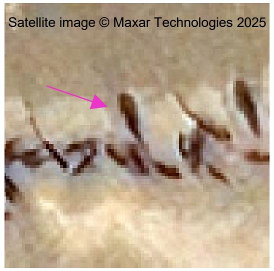
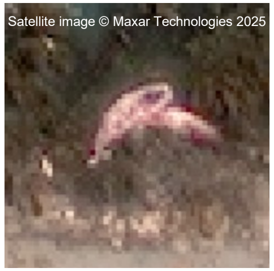
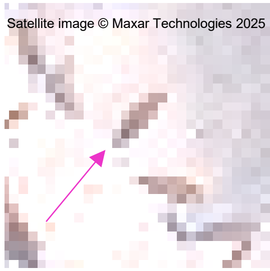
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
						
	passive_decomposition					 
	dry_remains			 		  

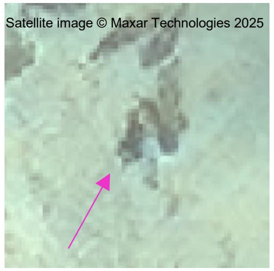
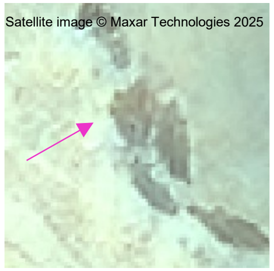
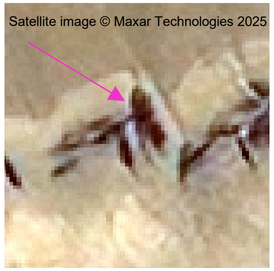
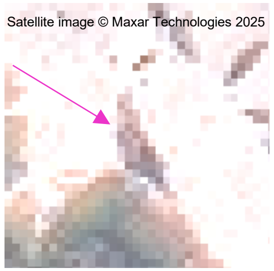
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
						
	unknown					
***species**	sei_whale *Balaenoptera borealis*					
	sperm_whale *Physeter macrocephalus*					

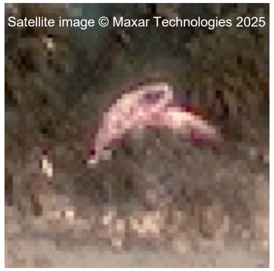
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	long-finned_pilot_whale *Globicephala melas edwardii*					
***colour**	black					
	brown					

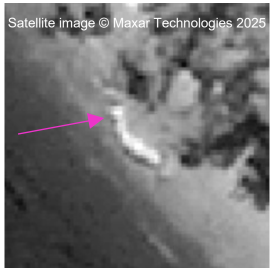
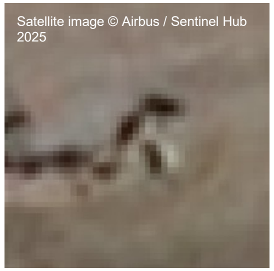
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	cream			 		
	grey_dark					
	grey_light					

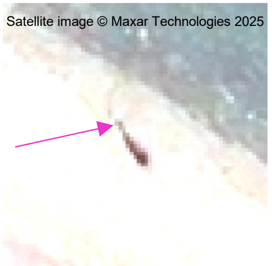
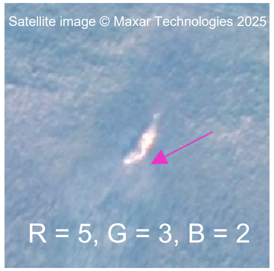
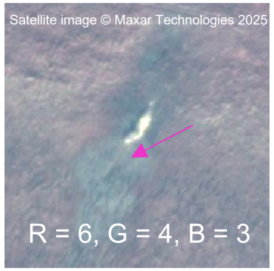
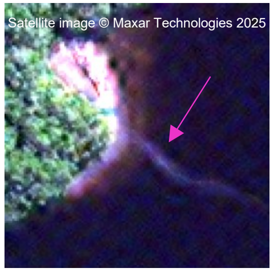
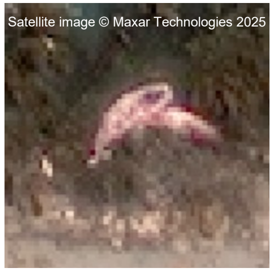
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	orange					
	pink					
	red					

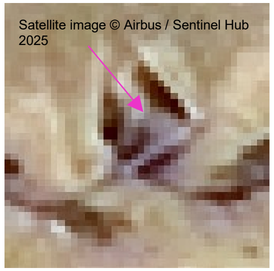
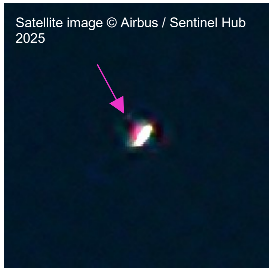
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	white		 			 
***body_shap**	ellipsoid		 	 	 	

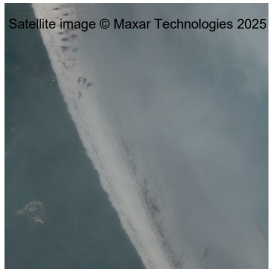
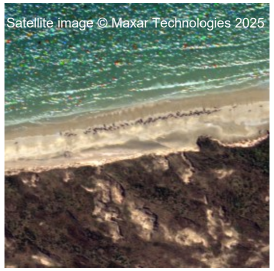
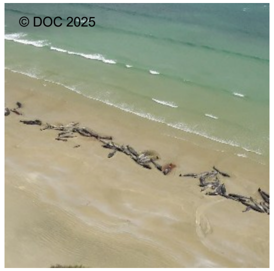
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	streamline					
	irregular_ellipsoid					
	irregular_streamline					
	undefinable					

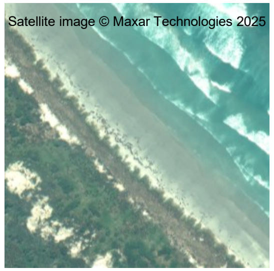
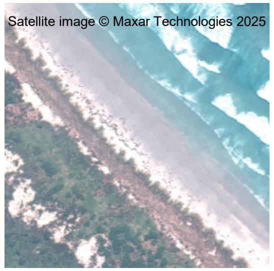
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
***flippers**	yes					 
	no					
	maybe					

Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
***fluke**	yes			 	 	 
	no					

Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	maybe					
***slick**	yes		 			
	no					

Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	maybe					
*group	individual					
	2_to_5					

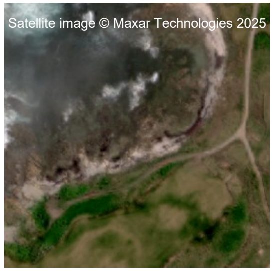
Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	6_to_10					
	11_to_50					
	51_to_100					
	101_to_200					

Attribute	Description	Satellite 0.15 m	Satellite 0.3 m	Satellite 0.5 m	Aerial	Ground
	201_to_500					
	501+					

****Please note: the following 'Fields' are environmental variables and requires the conditions at the point feature / annotation specifically, and not the whole image****

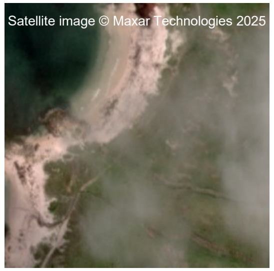
Cloud Cover (cloud_cov)

Table 3: Satellite image examples of the cloud cover values associated with the standardised attribute 'cloud_cov' that could impact carcass detection.

skc_(sky_clear)	few_(traces)	sct_(scattered)	bkn_(broken)	ovc_(overcast)
				

Cloud Thickness (cloud_thic)

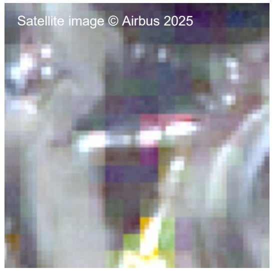
Table 4: Satellite image examples of the different cloud thickness values associated with the standardised attribute 'cloud_thic' that could impact carcass detection.

none	thin	medium_thin	thick
			

Environment (environmen)

Table 5: Satellite, aerial and ground image examples of the different abiotic environment values associated with the standardised attribute 'environmen' that a carcass may be found in.

	Satellite	Aerial	Ground
bedrock			
cobble_beach			
estuary_mud_flat			
gravel_beach			

man_made_structure			
mangrove			
marsh_brackish			
marsh_freshwater			
nearshore_waters_<1km			
offshore_waters_>1km			
sand_beach_dark			

sand_beach_light			
shell_beach			
vegetated_trees			
vegetated_wetland			

Image brightness (img_bright)

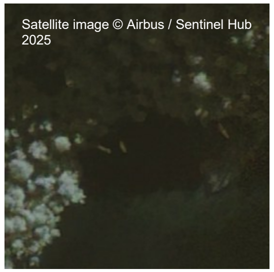
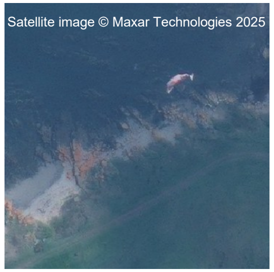
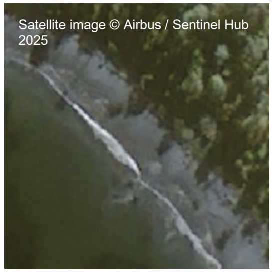
Table 6: Satellite image examples of the different image brightness values associated with the standardised attribute ‘img_bright’ that could impact carcass detection.

bright	**moderate**	**dark**

Shade (shade)

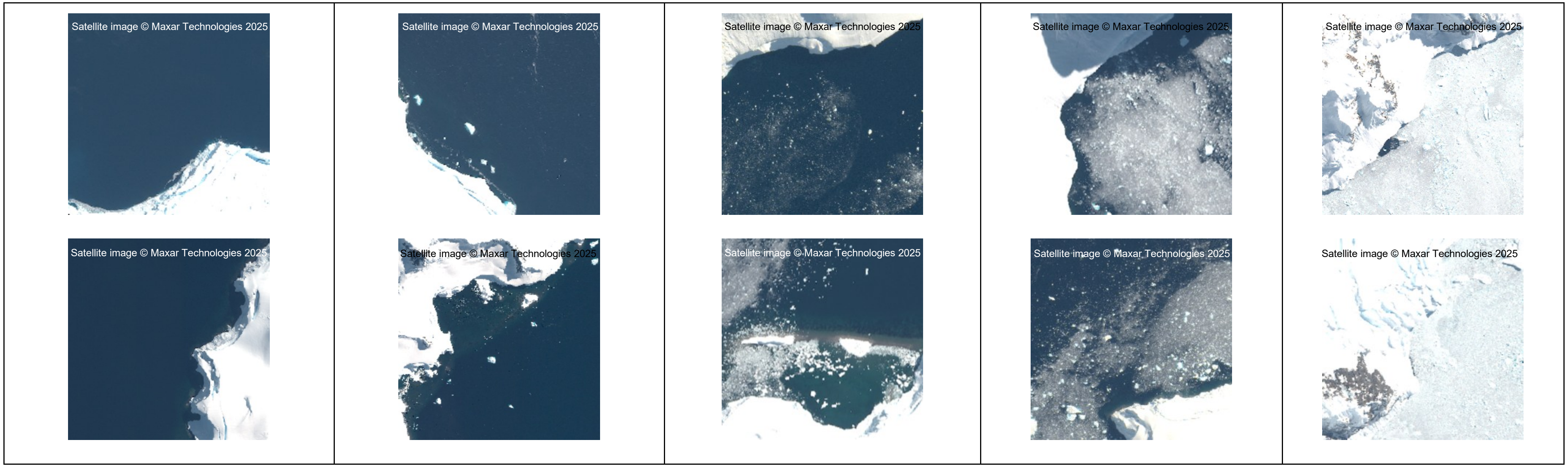
Table 7: Satellite image examples of the different shade values associated with the standardised attribute 'shade' that could impact carcass detection.

full_shade	**partial_shade**	**no_shade**
 		 

Ice_Cover (ice_cover)

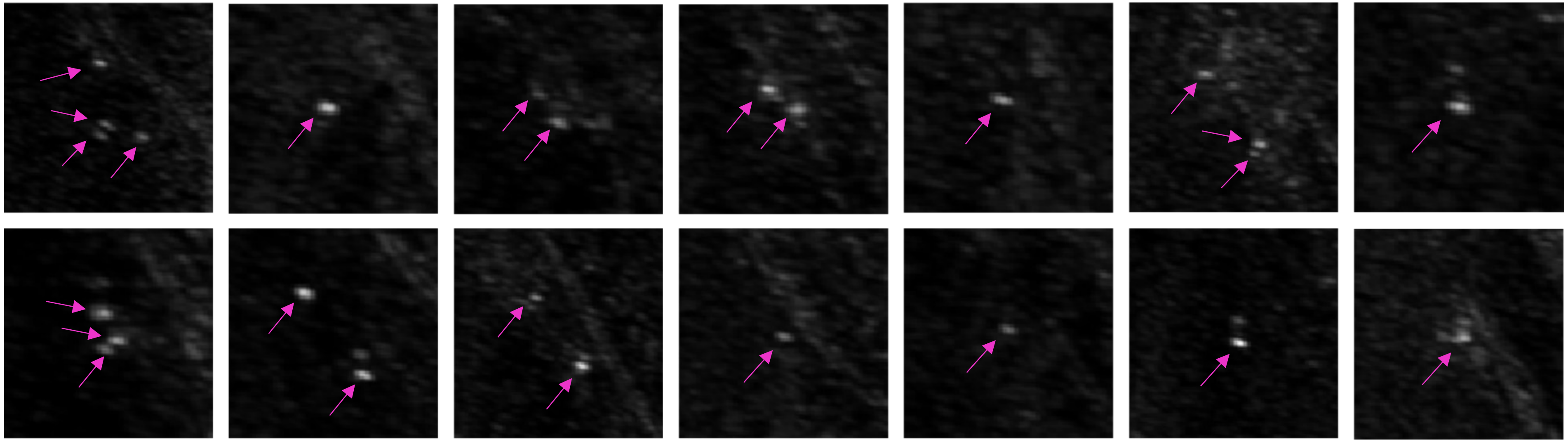
Table 8: Satellite image examples of the different ice cover values associated with the standardised attribute 'ice_cover' that could impact carcass detection. This attribute is only relevant when reviewing satellite imagery for regions where ice is present.

none	**low**	**medium**	**dense**	**very_dense**



This document is intended to guide the detection of stranded cetaceans in optical satellite imagery; however, below we include examples of stranded long finned pilot whale carcass detected in Synthetic Aperture Radar satellite imagery. Carcass change the surface roughness profile and increase the intensity of backscatter, to appear bright.

Figure 12: Examples of stranded pilot whales in Synthetic Aperture Radar imagery. Satellite imagery © DLR e.V. 2023, Distribution by Airbus Defence and Space GmbH



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